6. Transformations using matrices

What students need to learn:

Linear transformations of column vectors in two dimensions and their matrix representation.	The transformation represented by $\mathbf{AB}$ is the transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$.
Applications of 2×2 matrices to represent geometrical transformations.	Identification and use of the matrix representation of single transformations from: reflection in coordinate axes and lines $y = \pm x$, rotation through any angle about $(0, 0)$, stretches parallel to the x-axis and y-axis, and enlargement about centre $(0, 0)$, with scale factor k, $(k \neq 0)$, where $k \in \mathbb{R}$.
Combinations of transformations.	Identification and use of the matrix representation of combined transformations.
The inverse (when it exists) of a given transformation or combination of transformations.	Idea of the determinant as an area scale factor in transformations.